

Minimum resolving families in the six-element Jaccard space

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Abstract

The Jaccard metric on all subsets of a six-element set has metric dimension four. An elementary incidence-cube argument excludes every family of at most three landmarks. Exact exhaustive enumeration gives 34,860 minimum resolving families and classifies them into 70 orbits under ground-set permutations; we list every canonical representative. A separate enumeration of six-element subsets of the four-dimensional binary cube reconstructs the classification. The upper bound four was already reported by Lladser and Paradise. The contribution here is a matching lower bound and a complete finite classification, with reproducible certificates; priority beyond the inspected sources remains uncertain.

1 The finite problem and its context

Let $X = [6] = \{1, \dots, 6\}$ and let $\mathcal{P}(X)$ contain all 64 subsets, including $\emptyset$ and X . Set

$$d(A, B) = \text{Jac}(A, B) = \begin{cases} |A \triangle B|/|A \cup B|, & A \cup B \neq \emptyset, \\ 0, & A = B = \emptyset. \end{cases}$$

A family $R \subseteq \mathcal{P}(X)$ is *resolving* if the signatures $(d(A, r))_{r \in R}$ distinguish all points $A \in \mathcal{P}(X)$. Its landmarks are distinct and unordered; an ordering is used only to write signatures. The metric dimension β is the smallest possible $|R|$. We identify families only under the natural action of S_6 on X , without adjoining complementation to the equivalence relation.

Lladser and Paradise [1, Table 1] report the upper bound $\beta \leq 4$ at $|X| = 6$ using the information content heuristic. Their Proposition 1.1 gives necessary ground-element separation conditions. Kjos-Hanssen [2, Section 5, Table 1] lists the exact dimensions 1, 2, 2, 3, 3 for ground-set sizes one through five. The present problem is the next finite minimization together with its complete classification. It does not address the second-order asymptotic question in [2, Question 6.1].

Encode a subset A by the integer $m(A) = \sum_{i \in A} 2^{i-1}$, and encode a family by its increasing tuple of masks. Canonical representatives below are lexicographically least among all such tuples in their S_6 orbit.

Theorem 1. *The space $(\mathcal{P}([6]), d)$ has metric dimension four. Exactly 34,860 unordered four-element families resolve it. They form exactly 70 S_6 orbits, with the complete canonical representatives in Table 1. Four orbits have size 120, three have size 180, 32 have size 360, and 31 have size 720. In particular,*

$$R_* = \{\{1\}, \{2, 3\}, \{2, 4, 5\}, \{2, 4, 6\}\}, \quad m(R_*) = (1, 6, 26, 42),$$

is a minimum resolving family.

The lower bound below is elementary. The classification is an exact computer-assisted finite result. Its additional content over the cited upper bound is certified minimality and the catalogue of all minimum families. As broad adjacent context for exact finite certificates in discrete geometry, see Zeng et al. [3]; no theorem from that work is a premise here. The comparison with the Jaccard literature is confined to the cited versions and the bounded source inspection of 9 September 2026. It establishes neither worldwide priority nor the absence of other accounts of this finite result.

2 Minimality and an attaining family

For $k = |A|$, $s = |r|$ and $t = |A \cap r|$, one has

$$d(A, r) = \frac{k + s - 2t}{k + s - t} \quad \text{when } k + s - t > 0. \quad (1)$$

In particular, points of the same positive cardinality with equal intersection counts against all landmarks have identical signatures. This direct observation is also the fixed-cardinality criterion used in [2, Lemma 2.1].

Lemma 1. *No family of at most three landmarks resolves $\mathcal{P}([6])$.*

Proof. Suppose three landmarks r_1, r_2, r_3 resolve. Associate to each $x \in X$ the incidence column

$$c_x = (\mathbf{1}_{x \in r_1}, \mathbf{1}_{x \in r_2}, \mathbf{1}_{x \in r_3}) \in \{0, 1\}^3.$$

If $c_x = c_y$ for $x \neq y$, the singletons $\{x\}$ and $\{y\}$ have equal intersection counts and hence equal distances to every landmark by (1). Thus the six columns are distinct.

The eight cube vertices split into four disjoint opposite pairs $\{v, \mathbf{1} - v\}$. Omitting two vertices destroys at most two complete pairs. Hence the six columns contain two disjoint opposite pairs, say $\{c_a, c_b\}$ and $\{c_c, c_d\}$. Their sums both equal $(1, 1, 1)$. The distinct two-element points $A = \{a, b\}$ and $B = \{c, d\}$ consequently meet every landmark in exactly one element. Equation (1) again gives identical signatures, a contradiction. Finally, a smaller resolving family could be extended to three distinct landmarks, and adding coordinates preserves resolution. This excludes every smaller family as well. $\square$

To check attainment without rounding, multiply each distance by 60, the least common multiple of $1, \dots, 6$. Every nonzero union size divides 60, so these scaled distances are integers. Table 2 lists the signatures of all 64 points for the ordered landmarks $(1, 6, 26, 42)$, scaled in this way. Its rows are pairwise distinct. Thus R_* resolves and Lemma 1 proves $\beta = 4$.

An additional finite exclusion certificate accompanies the paper: for every family of size zero, one, two or three it supplies distinct point masks $a < b$ whose distances to all its landmarks agree. The $1 + 64 + 2016 + 41664 = 43,745$ rows cover those families exactly once in combination order. The certificate checker uses rational arithmetic on explicit subsets, including the specified empty-set convention.

3 Complete enumeration and the permutation classes

Let P be the set of all $\binom{64}{2} = 2016$ unordered pairs of distinct point masks. For each landmark r , form

$$E_r = \{\{a, b\} \in P : d(a, r) = d(b, r)\}.$$

Then R resolves if and only if $\bigcap_{r \in R} E_r = \emptyset$, where the intersection for the empty family is P . The enumerator represents each E_r by a bit mask. To construct it, it represents a distance

by $(|a \triangle r|, |a \cup r|)$, replacing $(0, 0)$ by $(0, 1)$, and compares two distances by integer cross multiplication. It visits every increasing k -tuple of masks for $0 \leq k \leq 4$, without symmetry pruning or early termination of the family search. The resulting exact counts are

k	Families tested $\binom{64}{k}$	Resolving families
0	1	0
1	64	0
2	2,016	0
3	41,664	0
4	635,376	34,860

This explicitly exhausts the unordered landmark domain. The complete list of successful quadruples is retained, not just the count.

A ground permutation preserves union and symmetric-difference cardinalities, so it preserves resolution. Starting from the set of successful quadruples, take its least remaining tuple, form all its images under the 720 ground permutations, and remove those images. Repeat until the set is empty. Every image is checked to lie in the remaining solution set, and the selected tuple is checked to be the least in its orbit. Therefore the extracted orbits are disjoint, exhaustive, and canonically represented. The number h of ground permutations fixing each unordered family is counted directly, and each orbit has size $720/h$. Table 1 records all of them; in particular,

$$4 \cdot 120 + 3 \cdot 180 + 32 \cdot 360 + 31 \cdot 720 = 34,860.$$

A separate complete reconstruction

Temporarily order four landmarks as the rows of a 4×6 binary matrix. Singleton separation forces its six columns to be distinct. Up to ground permutations, every such matrix therefore occurs by listing some six-element subset $C \subseteq \{0, 1\}^4$ in increasing column order. Search all $\binom{16}{6} = 8008$ such subsets, including those containing the zero column. Discard matrices with repeated rows, and otherwise check their 64 distance signatures using exact fractions computed from explicit sets. This gives 1162 resolving column sets.

Two such column sets represent equivalent unordered landmark families exactly when one is carried to the other by a permutation of the four coordinates: ground permutations reorder columns and landmark permutations reorder rows. Canonicalization under all 24 coordinate permutations gives 70 classes, agreeing individually with Table 1. If a coordinate permutation preserves C , the distinct columns determine a unique matching ground permutation. Conversely, a ground stabilizer of the family induces a unique row permutation because the rows are distinct. These correspondences identify the stabilizers, so a column class with stabilizer order h accounts for $720/h$ labeled families. This separately recovers every orbit size and the total $1162 \cdot 720/24 = 34,860$.

4 Reproducibility and limitations

The supplementary directory supplies two Python standard-library programs, the complete collision and solution lists, the orbit tables, and rational signatures for all 70 representatives. From that directory, with assertions enabled, the commands are

```
python3 evidence/enumerate_six.py
python3 evidence/verify_six.py
```

The first command regenerates the finite data. The second checks all 43,745 lower certificates and all 34,860 saved solutions, and performs the complete 8008-case column reconstruction.

These are deterministic integer or rational calculations; no floating-point tolerance or sampling enters the resolving predicate. Run the commands on a working copy of the supplement, since they regenerate its output files.

The classification has not been formalized in a proof assistant. An additional verification pass independently enumerated all quadruples by refining equal-signature classes of triples and reconstructed the S_6 orbits, agreeing with the full saved lists. This was a separate model-context check, not external peer review. The exact result here concerns only six ground elements; no value at a larger size or asymptotic refinement is asserted.

Computational and AI assistance. OpenAI Codex assisted with the mathematical argument, implementation of the exact searches, source comparison, and preparation of the manuscript. A separate Codex context performed the additional verification just described. The executable certificates specify the computational coverage; neither model output nor a successful compilation is used as a substitute for the arguments and finite checks.

References

- [1] M. E. Lladser and A. J. Paradise, *Metric Dimension and Resolvability of Jaccard Spaces*, arXiv:2405.11424v2 (2024). <https://arxiv.org/abs/2405.11424v2>.
- [2] B. Kjos-Hanssen, *The exact asymptotic constant in the metric dimension of Jaccard space*, arXiv:2609.09146v1 (2026). <https://arxiv.org/abs/2609.09146v1>.
- [3] Zijian Zeng, Houde Liu, Kuru Ratnavelu, and Ong Seng Huat, *Exact Finite Certificates for the Four-Dimensional Ternary Borsuk Problem and Five-Dimensional Ternary Kissing Codes*, Preprints (2026). <https://doi.org/10.20944/preprints202608.1658.v1>.

A The complete orbit catalogue

Each row represents the unordered family encoded by its four masks. The canonical convention is that of Section 1. Here o is the orbit size and $h = |\text{Stab}_{S_6}(R)|$, so $oh = 720$ in every row.

Table 1: All 70 minimum-family orbits under S_6 .

ID	Masks	o	h	ID	Masks	o	h
1	(1, 6, 26, 42)	360	2	36	(3, 13, 28, 53)	720	1
2	(1, 6, 26, 43)	720	1	37	(3, 13, 29, 53)	720	1
3	(1, 6, 27, 43)	360	2	38	(3, 13, 30, 52)	720	1
4	(1, 7, 26, 42)	360	2	39	(3, 13, 30, 53)	720	1
5	(1, 7, 26, 43)	720	1	40	(3, 13, 30, 54)	720	1
6	(1, 7, 27, 43)	360	2	41	(3, 13, 30, 55)	720	1
7	(3, 5, 24, 41)	360	2	42	(3, 13, 31, 53)	720	1
8	(3, 5, 25, 41)	180	4	43	(3, 28, 45, 53)	360	2
9	(3, 5, 25, 42)	720	1	44	(3, 29, 45, 53)	120	6
10	(3, 5, 25, 43)	720	1	45	(3, 29, 45, 55)	360	2
11	(3, 5, 25, 46)	360	2	46	(7, 11, 19, 45)	360	2
12	(3, 5, 25, 47)	360	2	47	(7, 11, 19, 61)	120	6
13	(3, 5, 26, 46)	720	1	48	(7, 11, 23, 49)	720	1
14	(3, 5, 30, 46)	180	4	49	(7, 11, 28, 45)	360	2
15	(3, 5, 30, 47)	360	2	50	(7, 11, 28, 49)	360	2
16	(3, 7, 25, 41)	360	2	51	(7, 11, 28, 53)	720	1
17	(3, 7, 25, 45)	720	1	52	(7, 11, 28, 61)	360	2
18	(3, 7, 29, 45)	360	2	53	(7, 11, 29, 45)	180	4
19	(3, 12, 21, 39)	720	1	54	(7, 11, 29, 47)	360	2
20	(3, 12, 21, 54)	720	1	55	(7, 11, 29, 49)	360	2
21	(3, 13, 21, 37)	120	6	56	(7, 11, 29, 50)	360	2
22	(3, 13, 21, 38)	360	2	57	(7, 11, 29, 53)	720	1
23	(3, 13, 21, 45)	720	1	58	(7, 11, 29, 62)	360	2
24	(3, 13, 21, 46)	720	1	59	(7, 11, 31, 49)	360	2
25	(3, 13, 21, 56)	360	2	60	(7, 15, 25, 43)	720	1
26	(3, 13, 21, 57)	360	2	61	(7, 25, 43, 45)	360	2
27	(3, 13, 21, 59)	360	2	62	(7, 25, 43, 46)	720	1
28	(3, 13, 22, 44)	720	1	63	(7, 25, 43, 55)	720	1
29	(3, 13, 22, 46)	720	1	64	(7, 27, 29, 43)	720	1
30	(3, 13, 22, 47)	720	1	65	(7, 27, 29, 46)	360	2
31	(3, 13, 22, 57)	720	1	66	(7, 27, 29, 47)	360	2
32	(3, 13, 23, 39)	360	2	67	(7, 27, 43, 53)	360	2
33	(3, 13, 23, 46)	720	1	68	(7, 27, 43, 57)	360	2
34	(3, 13, 23, 53)	720	1	69	(7, 27, 43, 61)	360	2
35	(3, 13, 23, 57)	720	1	70	(15, 23, 27, 61)	120	6

B An exact attainment certificate

The point masks run from 0 to 63. Each four-tuple in Table 2 is

$$60(d(A, 1), d(A, 6), d(A, 26), d(A, 42)),$$

using mask notation for landmarks. The scaling is exact, and no two tuples coincide.

Table 2: The 64 distinct integer signatures for R_* .

Point	Scaled signature	Point	Scaled signature
0	(60, 60, 60, 60)	32	(60, 60, 60, 40)
1	(0, 60, 60, 60)	33	(30, 60, 60, 45)
2	(60, 30, 40, 40)	34	(60, 40, 45, 20)
3	(30, 40, 45, 45)	35	(40, 45, 48, 30)
4	(60, 30, 60, 60)	36	(60, 40, 60, 45)
5	(30, 40, 60, 60)	37	(40, 45, 60, 48)
6	(60, 0, 45, 45)	38	(60, 20, 48, 30)
7	(40, 20, 48, 48)	39	(45, 30, 50, 36)
8	(60, 60, 40, 40)	40	(60, 60, 45, 20)
9	(30, 60, 45, 45)	41	(40, 60, 48, 30)
10	(60, 40, 20, 20)	42	(60, 45, 30, 0)
11	(40, 45, 30, 30)	43	(45, 48, 36, 15)
12	(60, 40, 45, 45)	44	(60, 45, 48, 30)
13	(40, 45, 48, 48)	45	(45, 48, 50, 36)
14	(60, 20, 30, 30)	46	(60, 30, 36, 15)
15	(45, 30, 36, 36)	47	(48, 36, 40, 24)
16	(60, 60, 40, 60)	48	(60, 60, 45, 45)
17	(30, 60, 45, 60)	49	(40, 60, 48, 48)
18	(60, 40, 20, 45)	50	(60, 45, 30, 30)
19	(40, 45, 30, 48)	51	(45, 48, 36, 36)
20	(60, 40, 45, 60)	52	(60, 45, 48, 48)
21	(40, 45, 48, 60)	53	(45, 48, 50, 50)
22	(60, 20, 30, 48)	54	(60, 30, 36, 36)
23	(45, 30, 36, 50)	55	(48, 36, 40, 40)
24	(60, 60, 20, 45)	56	(60, 60, 30, 30)
25	(40, 60, 30, 48)	57	(45, 60, 36, 36)
26	(60, 45, 0, 30)	58	(60, 48, 15, 15)
27	(45, 48, 15, 36)	59	(48, 50, 24, 24)
28	(60, 45, 30, 48)	60	(60, 48, 36, 36)
29	(45, 48, 36, 50)	61	(48, 50, 40, 40)
30	(60, 30, 15, 36)	62	(60, 36, 24, 24)
31	(48, 36, 24, 40)	63	(50, 40, 30, 30)